Proof. Assume there exists a least $k \in \mathbb{N}^+$, then it is clear that $\langle g \rangle := \{\dots g^{-2}, g^{-1}, e_G, g, g^2, \dots\} = \{e_G, g, g^2, \dots, g^{k-1}\}$, since $g^k = e_G$ and similarly, $g^{-k} = (g^k)^{-1} = e_G^{-1} = e_G$. Thus, for any $n \in \mathbb{Z}$, there exists $q \in \mathbb{Z}$ such that

$$g^n = g^{qk+\text{rem}(n,k)} = (g^k)^q \cdot g^{\text{rem}(n,k)} = g^{\text{rem}(n,k)}.$$

It follows that $|g| = |\langle g \rangle| \leq k$. To see that $|g| = k$, suppose by contradiction that $|g| < k$, then by the pigeonhole principle, there are two elements $0 \leq i < j < k-1$ such that $g^i = g^j$, hence $g^{j-i} = e_G$ and $j - i \geq 1$ and $j - i < k$, contradicting, k being the least such power.

It remains to show that no such k exists if $|g| = \infty$. Indeed, if $|g| < \infty$, then $g^m = e_G$ for some $m \in [1, |g|]$, so if no such m exists, $|g| = \infty$. ∎

Definition 0.0.1

A group G is called **cyclic** if $G = \langle g \rangle$ for some $g \in G$.

Example 0.0.2. $(\mathbb{Z}, +)$ is a cyclic group with 1 being a generator, as well as -1, since every integer is a multiple of 1 or of -1.

Example 0.0.3. $(\mathbb{Z}/n\mathbb{Z}, +)$ is a cyclic group of order n, where $\overline{1}$ is a generator. In fact, if $\gcd(k, n) = 1$, then $\overline{k}$ is also a generator.

Remark 0.0.4. We will show in the future that $\mathbb{Z}$, and $\mathbb{Z}/n\mathbb{Z}$ are the only cyclic groups up to an "isomorphism".

0.1 Subgroups Generated by a Subset

It is a fact (proven in the homework) that an intersection of subgroups is a subgroup itself. The proof also works for any arbitrary intersection of subgroups. The following remark follows from this fact:

Remark 0.1.5. Let G be a group and $S \subseteq G$, (S can be empty). There exists a smallest (with respect to inclusion) subgroup $H \leq G$ such that $H \supseteq S$ i.e. $H \supseteq S$ and for every $K \leq G$ and $K \supseteq S$, $H \subseteq K$. Namely,

$$H := \bigcap_{S \subseteq K \leq G} K.$$

Definition 0.1.6

Let G be a group and let $S \subseteq G$ (S can be empty). The **subgroup generated by** S, denoted $\langle S \rangle$ or $\langle S \rangle_G$, is the least subgroup containing S, i.e. the intersection of all subgroups containing S.

Example 0.1.7. For a group G, $g \in G$, the subgroup generated by g is the same as the subgroup generated by $\{g\}$.

Example 0.1.8. In $(\mathbb{Z}, +)$, we have that $\langle \{a, b\} \rangle = \gcd(a, b)\mathbb{Z}$ because by Theorem 6.1.2, $\gcd(a, b)$ can be written as a linear combination of a and b.

0.2 Permutation Groups

Recall that for a set X, we defined the group S_X to be all bijections from X to X. These bijections are called *permutations* of X and S_X is called the *group of permutations* or *permutation group* on X. In this case, we will mainly focus on permutation groups where X is finite. For these finite X, we define $S_n := S_X$ and call S_n the *permutation group of n elements*. We denote the permutations of S_n as $\sigma(1), \sigma(2), \ldots$. If $\sigma(i) = i$, we may omit it from the notation. A permutation of n elements can be written like so:

$$\begin{pmatrix} 1 & 2 & 3 & \ldots & n \\ \sigma(1) & \sigma(2) & \sigma(3) & \ldots & \sigma(n) \end{pmatrix}$$

where $1 \mapsto \sigma(1)$, $2 \mapsto \sigma(2)$ and so on. Note that since σ is a bijection, every element that appears in the top row will appear in the bottom row.

Example 0.2.9. Consider this composition in S_4:

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 3 & 4 \end{pmatrix}$$

To compute this permutation, we apply the innermost permutation, followed by the outermost one. Let $\sigma \in S_4$ be the outermost permutation and $\tau \in S_4$ be the innermost permutation. We see $\tau(1) = 2$, so $(\sigma \circ \tau)(1) = \sigma(\tau(1)) = \sigma(2) = 4$, so $1 \mapsto 4$. Similarly, $(\sigma \circ \tau)(2) = \sigma(\tau(2)) = \sigma(1) = 2$, so $2 \mapsto 2$, and so on. The composition $\sigma \circ \tau$ is thus:

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 3 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 2 & 3 & 1 \end{pmatrix}$$

1 October 21 Notes: Permutation Groups, Cyclic Permuations

Proposition 1.0.1

There are exactly $n!$ permutations of a set of cardinality n, i.e. $|S_n| = n!$.

Proof. Let $X = \{1, dots, n\}$ and $\sigma \in S_n$ be a permutation on X. Since σ is in bijection with $(\sigma(1), \ldots, \sigma(n))$, we have that $|S_n|$ is equal to the number of reorderings of $\{1, \ldots, n\}$. Thus the total number of possible reorderings is

$$n \cdot (n-1) \cdot (n-2) \cdot \cdots \cdot 1 = n!$$ ■

Definition 1.0.2

For a permutation $\sigma \in S_n$ we define its **support**, denoted $\text{supp}(\sigma)$, to be the set of all elements $\{1, \ldots, n\}$ which are not fixed by σ, i.e.

$$\text{supp}(\sigma) := \{k \in \{1, \ldots, n\} \sigma(k) \neq k\}.$$

We say two permutations $\sigma, \tau \in S_n$ are **disjoint** if $\text{supp}(\sigma) \cap \text{supp}(\tau) = \emptyset$.

Remark 1.0.3. Note that $\text{supp}(\sigma)$ is σ-invariant—that is—if $x \in \text{supp}(\sigma)$, then $\sigma(x) \in \text{supp}(\sigma)$.

Proposition 1.0.4

If $\sigma, \tau \in S_n$ have disjoint supports, then $\sigma \circ \tau = \tau \circ \sigma$, i.e. the permutations commute.

Proof. Let $i \in \{1, \dots, n\}$. If $i \notin \text{supp}(\sigma) \cup \text{supp}(\tau)$, then $(\sigma \circ \tau)(i) = \sigma(\tau(i)) = \sigma(i) = i$ and similarly, $(\tau \circ \sigma)(i) = \tau(\sigma(i)) = \tau(i) = i$. If $i \in \text{supp}(\sigma)$, then $i \notin \text{supp}(\tau)$, so $(\sigma \circ \tau)(i) = \sigma(\tau(i)) = \sigma(i)$ and $(\tau \circ \sigma)(i) = \tau(\sigma(i)) = i$ by Remark X.X.X, hence $\sigma(i) \notin \text{supp}(\tau)$. The case where $i \in \text{supp}(\tau)$ follows analogously. ■

Example 1.0.5. Let X be a set and $Y \subseteq X$. Define two subgroups of Y:

1. $F_Y := \{\sigma \in S_X : Y \cap \text{supp}(\sigma) = \emptyset\} = \{\sigma \in S_X : \forall y \in Y, \sigma(y) = y\}$. Clearly F_Y is closed under composition, inverses, and contains Id_X.
2. $I_Y := \{\sigma \in S_X : \sigma Y = Y\}$. This is also clearly a subgroup of S_X.

Remark that $F_Y \leq I_Y \leq S_X$, however if $|Y| > 1$, then $F_y \not\leq I_Y$.

Definition 1.0.6

A permutation $\sigma \in S_n$ is called a **cycle** if it permutes its support "cyclically," i.e. we can order $\text{supp}(\sigma) = \{a_0, a_1, \dots, a_{l-1}\}$ such that $\sigma(a_i) = a_{i+1 \pmod l}$ for all $0 \leq i < l$. We write a cyclic permutation σ as:

$$\sigma = (a_0, a_1, \dots, a_{l-1})$$

where l is the **length** of the cycle.

If $l = 0$, σ is called the *trivial cycle*, where $\sigma = \text{Id}_n$.

Remark 1.0.7. For any cycle $\sigma \in S_n$,

$$(a_0, a_1, \dots, a_{l-1}) = (a_1, \dots, a_{l-1}, a_0) = (a_i, a_{l-1}, , a_0, \dots, a_{i-1}).$$